TeachLens Test Recording Scenarios

Scripts for short simulated lectures with 4-6 students, filmed with two static cameras, to test the TeachLens pose pipeline

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What this booklet is for

Each scenario is a 2-5 minute scripted take that mimics a real lesson: an instructor who teaches well or badly, and students who are attentive, distracted, or somewhere in between. Every page says what the instructor does, what each student does and when, and what to check in the TeachLens outputs afterwards.

The scripted times and the ground-truth log you keep are what make these videos useful: they are the "right answers" that the model's results are compared against.

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1. Read me first

What TeachLens measures today, and what it does not

The current pipeline (List A) reports **observable physical evidence**: whether the instructor is visible, how many students are visible, anonymous tracks, sitting or standing, how much each person moves, apparent entries and exits at the frame edge, visibility and occlusion, and (once calibrated) classroom zones.

It does **not** judge teaching quality or student attention, and it does not yet detect phones, hand raises, gaze, sleeping, or talking. The scenarios still script those behaviours on purpose: your log of them is the ground truth for testing List A now, and for building and testing future indicators (hand raise, head orientation, objects) on the same videos. Labels such as "effective" and "attentive" describe the acting, not a TeachLens output.

The five rules for every take

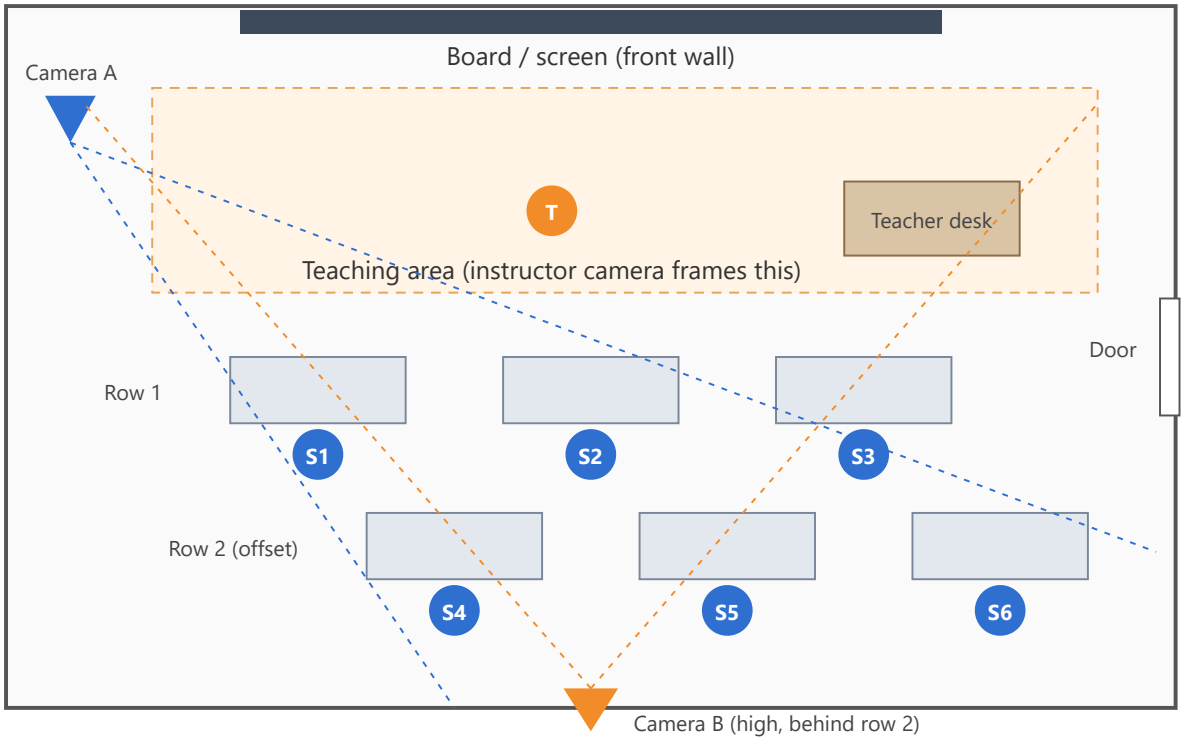
1. **Both cameras stay still.** Tripods, no panning, no zooming, no refocusing during or between the takes of a session. TeachLens now assumes static cameras.
2. **Each camera sees only its own people.** The student camera shows only students, and the instructor camera shows only the instructor, except where a scenario says otherwise (B7, C5, D2).
3. **Start every take the same way.** Start both cameras, clap once where both cameras can see it, hold up the take sign (for example "A1 take 2", never names), then everyone sits still for 15 s before the action starts.
4. **Students stand up in place.** Student legs are hidden by the desks and TeachLens does not use them. Sitting or standing is judged from how far a student's shoulders rise above their own seated level, so stand straight up behind the desk, keep both shoulders visible, and stay seated for at least the first 15 s.
5. **Log what really happened.** Acting never matches the script exactly. The logger writes down the actual times (section 6), and that log, not the script, is the ground truth.

How the scenarios are organised

- **A. Teaching quality and class attention** (A1-A6): the core matrix, including attention that declines or recovers over time.
- **B. Instructor movement and posture** (B1-B8): standing, walking, pacing, sitting, board writing, leaving, roaming, and legs cut off by the frame.
- **C. Student events** (C1-C6): standing answers, arrivals and departures, pair work, silent work, presentations, and video watching.
- **D. Stress tests** (D1-D2), **N. a natural unscripted lesson** (N1), and **K. calibration clips** (K1-K3).

Every scenario mixes behaviours the way a real class does: even an "attentive class" has a brief phone check, and even an "inattentive class" has one student taking notes. Priority 1 scenarios are the minimum useful set; film them first.

2. Camera and room setup



Camera A: student camera (`student_view`)

- At the front of the room, beside the board, facing the students.
- About 1.8-2.2 m high, tilted slightly down, so every head and both shoulders of every student are clear.
- Frame only the students. The instructor walking at the board is behind this camera.
- The door should be at a side of the frame, so entries and exits cross the left or right edge.

Camera B: instructor camera (`instructor_view`)

- Behind the back row, raised above the students' heads, facing the board.
- Frame the whole teaching area with **no student heads** in the picture: a second person makes the camera's role rule report "unknown".
- Prefer framing the instructor's whole body at the board. Legs leaving the bottom of the frame near the camera is fine; TeachLens handles it (scenario B6 tests it).

Camera settings (both cameras)

- Landscape, 1080p, 30 fps (constant frame rate if the app offers it). Lock exposure and focus (on phones, long-press to lock AE/AF).
- Turn off auto-zoom, auto-framing ("Center Stage" and similar), stabilisation cropping, and HDR if possible. TeachLens treats every image as the same scene, so the picture must not shift.
- Do not touch the camera during a session. If a camera is bumped or moved, note the time and record K1 again (zones depend on the exact view).
- Even lighting; avoid bright windows behind the students (the pose model loses joints against glare). Keep the lights the same throughout a take.

Setup check (before every session)

1. Everyone takes their seat; record 10 s on both cameras and look at a frame.
2. Student camera: both shoulders of every student are visible; back-row students are visible between front-row students (seats offset); the instructor is not in the frame.
3. Instructor camera: the whole teaching area is in the frame and no student heads are visible.
4. Check that the clap is visible to both cameras.

3. Cast, seating, and acting guide

Seats

Seats are numbered as seen from the student camera: **row 1** (front) S1 left, S2 middle, S3 right; **row 2** (back, offset half a seat to the right) S4, S5, S6. Every scenario is written for six students and says how to run it with four or five: usually drop S6, then S5. The roles of S1-S4 are written so the scenario keeps its balance with four students.

Wear plain tops in clearly different colours. This only helps the human logger tell students apart; TeachLens identifies nobody and uses anonymous tracker numbers.

Rotate roles between takes

For a second take, swap the roles of S1 and S3, and of S4 and S6 (S2 and S5 keep theirs). Behaviours then do not always come from the same person or seat, and neighbouring pairs stay neighbours. Record the rotation in the log.

Acting the instructor

Effective teaching looks like	Ineffective teaching looks like
Faces the class and makes eye contact; uses the board while explaining; purposeful walking and gestures; asks questions and waits; picks students who raise hands; checks understanding.	Reads notes or slides aloud in a monotone; long stretches with the back turned; sits the whole time; ignores raised hands; no questions; aimless or nervous pacing; checks own phone.

Acting the students

- Make every behaviour **clear but natural**: a phone should be visible in the hand, a chat should turn head and shoulders, sleeping means the head down on the arms.
- Hold each behaviour for its full time window; do not drift between behaviours unless the script says so.
- Raise a hand by lifting the arm clearly above the head and holding it 3-5 s, while staying seated.
- Stand straight up in place, hold, then sit fully. Step into the aisle only when the script says the student leaves the seat.
- Off-task acting stays safe and kind: nothing thrown except a soft paper ball in D2.

Lecture topics (any of these, reused across takes)

- **The water cycle**: draw evaporation, condensation, precipitation, and collection as a loop.
- **Counting in binary**: write 0-8 in binary and explain place values.
- **Supply and demand**: draw the two curves and the equilibrium point.

> ### Consent and storage
>
> Get written consent from every person filmed (recording, and use for SGP testing and in the project report). Use anonymous take IDs; no names in file names, on take signs, or on screens. Keep the videos local and never commit them: `data/raw_videos/` is ignored by Git. Note that the TeachLens folder is inside OneDrive, so files in it sync to the cloud; keep the recordings in a non-synced folder if your consent terms promise local-only storage. Record each video in `docs/selected_videos.md`.

4. Behaviour codes

Scripts and the log use these codes. "TeachLens today" says what the current pipeline can observe for that behaviour; columns marked later are future indicators that the same videos will help build.

Instructor

Code	Behaviour	How to act it	TeachLens today
T-STAND	Standing, facing the class	Stand at the front, weight on both feet, face the class, talk with eye contact.	Presence; posture standing; motion still/low.
T-GEST	Explaining with gestures	Feet still. Point at the board, use open-hand gestures, turn the torso between board and class.	Motion low/medium without walking.
T-WALK	Walking while teaching	Walk at a normal speaking pace across the front while talking; pause 3-5 s at each end.	Motion low/medium; hip path with `--path teacher`.
T-PACE	Nervous pacing	Walk quickly back and forth with frequent turns and no pauses.	Motion medium/high; posture often unknown mid-stride.
T-WRITE	Writing on the board	Back to the class; write for the stated time and talk to the board.	Presence; face not visible; posture standing.
T-READ	Reading from notes	Look down at notes, a laptop, or a phone; read aloud in a monotone; rarely look up.	Still motion. Head-down reading is not measured yet.
T-SIT	Seated at the desk	Sit in the teacher's chair behind the desk (the desk hides the legs).	Posture sitting or unknown (legs hidden); still.
T-LEAN	Perched on the desk	Half-sit on the front edge of the desk, arms crossed or resting.	Posture usually unknown (ambiguous); still.
T-ASK	Asking the class	Ask a question, wait 3-5 s, then point to a student.	Motion low (pointing). Questions themselves are audio (out of scope).
T-NEAR	Close to the instructor camera	Walk toward the instructor camera until the knees, then the hips, leave the bottom of the frame.	Legs cut off: posture partial or unknown, and no false exit.
T-ROAM	Walking among the students	Leave the teaching area, walk along the aisle, stop at desks.	Instructor view: not detected (out of view, not absent). Student view: counted as a student (known limitation).
T-EXIT	Leaving the room	Walk out through the door, crossing the side of the frame; come back later.	Apparent exit/entry at the frame edge; presence not detected while away.
T-SCREEN	Beside the screen	Stand still at the side, next to a screen or TV playing a video.	Standing; still.

Students

Code	Behaviour	How to act it	TeachLens today	Later
S-ATT	Attentive	Sit upright facing the instructor or board, follow the instructor with your head, nod now and then.	Sitting; still/low motion.	Head orientation
S-NOTE	Taking notes	Head down writing; look up at the board every 10-20 s.	Sitting; low motion.	Note-taking vs. phone
S-READ	Reading course material	Read a handout or textbook on the desk and turn pages.	Sitting; low motion.	Object (book)
S-HAND	Raising a hand	Raise one arm clearly above the head, hold 3-5 s, lower it. Stay seated.	Must stay **sitting**: a hand raise is not standing.	Hand raise (List C)
S-STAND	Standing up to answer	Stand straight up **in place** behind the desk, hold 5-8 s, sit down. Do not step sideways.	**Standing** from shoulder rise.	-
S-GROUP	On-task pair work	Turn toward the partner, point at a shared paper, talk about the task.	Sitting; low/medium motion; overlap.	Interaction
S-PHONE	Phone use	Phone in the lap or on the desk, head down, thumb scrolling; glance up rarely.	Sitting; still. Not yet told apart from notes.	Phone + head orientation
S-LAPTOP	Off-task laptop	Look at the laptop screen, typing or scrolling unrelated content.	Sitting; still/low.	Object (laptop)
S-TALK	Off-task chatting	Turn head and shoulders to a neighbour, chat and laugh quietly.	Sitting; low/medium; turning can hide a shoulder (posture unknown).	Orientation, interaction
S-SLEEP	Sleeping	Fold the arms on the desk and rest the head on them; stay down.	Sitting (shoulders drop below the seated level); still.	Head-down duration
S-SLOUCH	Disengaged slouch	Lean far back, arms crossed, look at the ceiling or away.	Sitting or unknown; still.	Posture/orientation
S-LOOK	Looking away	Look out of the window, at the door, or to the back of the room for 10 s or more.	Sitting; still.	Head orientation
S-FIDGET	Restless	Shift in the seat, stretch the arms, turn around, tap a pen.	Sitting; motion bursts.	-
S-EAT	Eating or drinking	Eat a snack or drink from a bottle.	Sitting; low motion.	Object
S-LEAVE	Leaving the room	Stand, walk to the door, and leave through the side of the frame.	Count drops; apparent exit; posture unknown away from the seat.	-
S-LATE	Late arrival	Walk in through the door, go to the seat, sit down.	Count rises; apparent entry; new tracker ID.	-

5. Scenario overview and filming plan

ID	Scenario	Teaching	Class	Length	Prio.
A1	Effective teaching, attentive class	Effective	Attentive (5 of 6)	3:00	1
A2	Effective teaching, inattentive class	Effective	Inattentive (1 of 6)	3:00	1
A3	Ineffective teaching, attentive class	Ineffective	Attentive (diligent)	3:00	1
A4	Ineffective teaching, inattentive class	Ineffective	Inattentive (1 of 6)	3:00	1
A5	Attention declines over time	Effective, then ineffective	Declining (6 to 1)	4:00	1
A6	Instructor wins the class back	Ineffective, then effective	Recovers	3:30	2
B1	Standing-still lecture	Effective (static)	Mostly attentive	3:00	2
B2	Walking lecture with a side exit	Effective (mobile)	Mostly attentive	3:00	1
B3	Pacing and large gestures	Nervous / over-animated	Mostly attentive	2:30	2
B4	Seated instructor and posture changes	Mixed	Mostly attentive	3:00	1
B5	Board-writing lesson	Mixed	Copying; one opportunist	3:00	2
B6	Near and far: legs leaving the frame	n/a (camera test)	Attentive	2:30	1
B7	Instructor walks among the students	Effective (circulating)	Working	3:00	2
B8	Instructor leaves the room	n/a	Off task while unsupervised	3:00	1
C1	Question time: hands and standing answers	Effective (interactive)	Attentive	3:30	1
C2	Late arrival, restroom break, early leave	Effective	Mostly attentive	3:30	1
C3	Pair work: on task and off task	Effective (activity)	Mixed (4 on, 2 off)	3:00	2
C4	Silent individual work (quiz)	n/a (supervision)	On task, heads down	3:00	2
C5	Student presentation (two people in front)	n/a	Attentive	3:00	2
C6	Watching a video	n/a (media)	Mixed	3:00	2
D1	Occlusion and crowding	Effective	Mostly attentive	2:30	2
D2	Off-task disturbance and recovery	Effective (manages the class)	Disrupted, then recovers	3:00	2
N1	Natural, unscripted mini-lesson	Natural	Natural	5:00	1
K1	Empty room	n/a	n/a	0:30	1
K2	Student sit/stand calibration	n/a	n/a	2:30	1
K3	Instructor posture and position grid	n/a	n/a	2:30	1

Suggested sessions

Session	Scenarios
Session 1 (about 2 h)	Setup check, K1, K2, K3, A1, A2, A3, A4 (two takes each for A1-A4)
Session 2 (about 2 h)	A5, C1, C2, B2, B4 (two takes: desk and open chair), B6, B8, N1
Session 3 (about 2 h)	A6, B1, B3, B5, B7, C3, C4, C5, C6, D1, D2, and a second N1

Film two takes of every Priority 1 scenario (roles rotated in take 2) and one take of each Priority 2 scenario. Running order inside a session does not matter, but record K1 first and again whenever a camera moves.

What each scenario exercises

Dots mark the indicators a scenario is designed to test. Columns with * are not measured by TeachLens yet; those scenarios provide ground truth for future indicators. "S posture" is student sitting/standing from shoulder rise; "Role rule" is the camera-based role assumption (one person in the instructor camera; students only in the student camera).

Scenario	Presence	Count	Tracking	T posture	S posture	Motion	Entry / exit	Occlusion	Role rule	Legs cut off	Hand raise*	Attention*	Phone*
A1 Effective teaching, attentive class	●	●		●	●	●					●	●	●
A2 Effective teaching, inattentive class	●	●				●					●	●	●
A3 Ineffective teaching, attentive class	●			●		●					●	●	
A4 Ineffective teaching, inattentive class	●			●		●						●	●
A5 Attention declines over time	●					●					●	●	●
A6 Instructor wins the class back				●	●	●					●	●	●
B1 Standing-still lecture	●			●		●							
B2 Walking lecture with a side exit	●		●			●	●						
B3 Pacing and large gestures				●		●							
B4 Seated instructor and posture changes				●		●							
B5 Board-writing lesson	●					●						●	●
B6 Near and far: legs leaving the frame				●			●			●			
B7 Instructor walks among the students	●	●							●		●		
B8 Instructor leaves the room	●		●		●		●					●	●
C1 Question time: hands and standing answers					●						●		
C2 Late arrival, restroom break, early leave		●	●				●						
C3 Pair work: on task and off task					●			●				●	●
C4 Silent individual work (quiz)					●	●					●	●	
C5 Student presentation (two people in front)	●	●							●		●		
C6 Watching a video						●						●	●
D1 Occlusion and crowding			●		●			●					
D2 Off-task disturbance and recovery	●	●				●			●			●	●
N1 Natural, unscripted mini-lesson	●	●	●	●	●	●						●	
K1 Empty room	●	●											
K2 Student sit/stand calibration					●						●		
K3 Instructor posture and position grid				●						●			

Teaching quality and class attention. The core matrix: effective or ineffective teaching, crossed with attentive, inattentive, declining, and recovering classes.

A1 Effective teaching, attentive class

Priority 1: film first | Teaching: Effective | Class: Attentive (5 of 6) | Length: 3:00

Purpose. Baseline good lesson: a confident instructor who stands, gestures, and walks; most students attentive, with one brief realistic distraction and one standing answer.

Cast. 6 students. With 4, drop S5 and S6; with 5, drop S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-0:45	T-STAND	Introduce the topic from the centre of the front, eye contact, open gestures.
0:45-1:15	T-WRITE, T-GEST	Write a short heading (no more than 10 s), then explain the diagram, facing the class half the time.
1:15-1:30	T-WALK	Walk slowly to the left side while talking.
1:30-1:50	T-ASK	Ask a question; pick S3, who answers standing.
1:50-2:30	T-WALK, T-GEST	Walk back to the centre and summarise with gestures.
2:30-3:00	T-STAND	Ask "any questions?", answer S6, close.

Students

Seat	Role	What to do (codes and times)
S1	Attentive	S-ATT throughout; S-NOTE 1:00-1:20.
S2	Note-taker	S-NOTE throughout (looks up every 10-15 s).
S3	Answers	S-ATT; S-HAND at 1:30 (4 s); **S-STAND 1:35-1:45** to answer; then S-ATT.
S4	Brief distraction	S-ATT; **S-PHONE 2:05-2:20** (checks a message), puts it away, S-ATT.
S5	Note-taker	S-NOTE; S-HAND at 1:30 together with S3, lowers when S3 is picked.
S6	Asks at the end	S-ATT; S-HAND at 2:40 and asks a question while seated.

Check in the TeachLens outputs

- Instructor camera: presence visible almost all the time; posture standing (full or partial legs); motion still/low when standing, low/medium while walking; no apparent entries or exits.
- Student camera: count equals the cast after about 1 s; S3 standing about 1:35-1:45; everyone else sitting (or unknown for the first 3 s of each track).
- Hand raises at 1:30 and 2:40 stay **sitting**.

Watch for

- S3 must stand straight up in place, not step into the aisle.
- The instructor stays out of the student camera's view.

A2 Effective teaching, inattentive class

Priority 1: film first | Teaching: Effective | Class: Inattentive (1 of 6) | Length: 3:00

Purpose. Same strong delivery as A1, but the class does not respond. Shows that instructor-camera evidence cannot tell A1 from A2: attention differences live only in the student camera and the ground-truth log.

Cast. 6 students. With 4, drop S5 and S6 (still 1 of 4 attentive).

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-0:50	T-STAND, T-GEST	Energetic introduction with gestures.
0:50-1:30	T-WALK	Walk left and right while explaining, looking at the students.
1:30-1:45	T-ASK	Ask a question, wait 5 s, no hands; ask again; S1 answers seated.
1:45-2:30	T-WRITE, T-GEST	Write briefly, then explain; step toward the front edge of the teaching area.
2:30-3:00	T-STAND	Try an attention grabber (a joke or "this is on the exam"), wrap up.

Students

Seat	Role	What to do (codes and times)
S1	The engaged one	S-ATT and S-NOTE throughout; answers seated at about 1:40.
S2	Phone	S-PHONE throughout; glances up about every 30 s.
S3	Asleep	**S-SLEEP 0:30-2:20**, then S-SLOUCH.
S4	Bored	S-SLOUCH and S-LOOK; **S-FIDGET 1:00-1:30** (turns around, stretches).
S5	Chatting	S-TALK with S6 from 0:40; S-PHONE from 2:00.
S6	Drifts off	S-ATT until 0:40, then S-TALK with S5; S-LAPTOP from 2:30.

Check in the TeachLens outputs

- Instructor camera: similar to A1 (presence, standing, low/medium motion).
- Student camera: everyone sitting; S3's shoulders drop below the seated level while asleep (shoulder rise negative, still sitting); motion bursts for S4 at 1:00-1:30.
- Phone, sleep, and chatting are **not measured yet**. Log them as ground truth for future head-orientation and object indicators.

Watch for

- Keep phones visible to the student camera (on the desk or lap, not hidden in a bag).
- Chatting students turn head and shoulders, not just the eyes.

A3 Ineffective teaching, attentive class

Priority 1: film first | Teaching: Ineffective | Class: Attentive (diligent) | Length: 3:00

Purpose. A dull lesson (reading aloud, long board writing, an ignored hand) to a diligent class, e.g. an exam next week. Tests a still instructor and a calm class.

Cast. 6 students. With 4, drop S5 and S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-1:15	T-READ	Stand behind the teacher's desk reading notes aloud, head down, monotone, almost no gestures.
1:15-2:15	T-WRITE	Long board writing with the back to the class, talking to the board.
2:00	-	Ignore S5's raised hand; keep writing.
2:15-3:00	T-READ	Back at the desk reading; end abruptly without questions.

Students

Seat	Role	What to do (codes and times)
S1	Copies everything	S-NOTE throughout.
S2	Upright listener	S-ATT; S-NOTE from 2:30.
S3	Tired but working	S-NOTE; S-FIDGET 1:45-1:55 (yawn, stretch), back to notes.
S4	Reader	S-READ the handout; looks up at the board while the instructor writes.
S5	Ignored question	S-ATT; **S-HAND 2:00-2:08** (ignored), lowers, S-NOTE.
S6	Note-taker	S-NOTE; S-LOOK briefly at 2:40 (5 s).

Check in the TeachLens outputs

- Instructor camera: presence visible; motion mostly still/low; posture unknown or standing (partial) behind the desk, standing at the board; face not visible during T-WRITE.
- Student camera: all sitting; low motion; the hand raise stays sitting.

Watch for

- The desk should hide the instructor's legs during T-READ: this tests posture with hidden legs.

A4 Ineffective teaching, inattentive class

Priority 1: film first | Teaching: Ineffective | Class: Inattentive (1 of 6) | Length: 3:00

Purpose. Worst case: a seated, disengaged instructor and a class that drifts off. Contrasts with A1 on both cameras.

Cast. 6 students. With 4, drop S5 and S6 (S4 is the one attentive student).

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-2:00	T-SIT, T-READ	Sit at the desk reading slides from a laptop; no eye contact.
2:00-2:20	T-READ	Check your own phone while the students "work".
2:20-2:45	T-STAND	Stand up behind the desk, say "read chapter 4".
2:45-3:00	T-SIT	Sit down again.

Students

Seat	Role	What to do (codes and times)
S1	Loses interest	S-ATT 0:15-1:00, then S-PHONE.
S2	Asleep	S-SLEEP from 0:40.
S3	Phone	S-PHONE throughout.
S4	The diligent one	S-NOTE throughout.
S5	Chatting	S-TALK with S6 from 0:30; S-EAT from 2:10.
S6	Chatting	S-TALK with S5 from 0:30; S-LAPTOP from 1:30.

Check in the TeachLens outputs

- Instructor camera: posture sitting or unknown (legs behind the desk; true posture is sitting); a stand-up at 2:20 and a sit-down at 2:45; motion still.
- Student camera: everyone sitting; little motion.

Watch for

- Record the exact stand-up and sit-down moments in the log; they are posture ground truth.

A5 Attention declines over time

Priority 1: film first | Teaching: Effective, then ineffective | Class: Declining (6 to 1) | Length: 4:00

Purpose. Retention decay: an engaging start that becomes monotone, with students drifting off at staggered times. Tests indicators as a timeline rather than one number.

Cast. 6 students. With 4, drop S5 and S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-1:15	T-STAND, T-GEST, T-WALK	Engaging start: gestures, walks, asks one question (S1 answers seated).
1:15-3:00	T-READ	Gradually stop moving; read from notes at the desk or the board.
3:00-3:30	T-WRITE	Back to the class, writing.
3:30-4:00	T-ASK	Ask "any questions?"; only S1 raises a hand.

Students

Seat	Role	What to do (codes and times)
S1	Persistent	S-NOTE until the end; S-HAND at 3:35.
S2	Drifts 1st	S-ATT; S-LOOK from 1:15; S-PHONE from 2:00.
S3	Drifts 2nd	S-ATT; S-SLOUCH from 1:30; S-SLEEP from 2:30.
S4	Drifts 3rd	S-ATT; S-FIDGET from 1:45; S-PHONE from 2:15.
S5	Drifts 4th	S-NOTE; S-TALK with S6 from 2:00.
S6	Drifts 4th	S-ATT; S-TALK with S5 from 2:00; S-LAPTOP from 3:00.

Check in the TeachLens outputs

- Instructor camera: motion_level timeline goes from low/medium to still; presence visible throughout.
- Student camera: all sitting; S3's shoulder rise goes negative after 2:30 (asleep), still sitting.
- Log the true number of attentive students per 30 s (6, 6, 5, 4, 2, 1, ...).

Watch for

- Make each change clear and keep it: drifting students should not snap back to attention.

A6 Instructor wins the class back

Priority 2 | Teaching: Ineffective, then effective | Class: Recovers | Length: 3:30

Purpose. The reverse of A5: the instructor notices disengagement, stands up, moves closer, asks questions, and most students re-engage.

Cast. 6 students. With 4, drop S5 and S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-1:00	T-SIT, T-READ	Read from the laptop at the desk.
1:00-1:10	T-STAND, T-WALK	Stop, stand up, walk to the front centre.
1:10-1:40	T-ASK	Ask direct questions; ask for hands.
1:40-2:30	T-GEST, T-WALK	Interactive explanation, walking.
2:30-2:45	T-ASK	Ask S2 to stand and answer.
2:45-3:30	T-STAND	Summarise.

Students

Seat	Role	What to do (codes and times)
S1	Steady	S-ATT throughout; S-HAND at 1:15.
S2	Re-engages	S-PHONE until 1:05, then S-ATT; **S-STAND 2:30-2:40** to answer.
S3	Relapses	S-SLOUCH until 1:05, S-ATT, then S-PHONE again from 2:45.
S4	Steady	S-NOTE throughout; S-HAND at 1:15.
S5	Re-engages	S-TALK with S6 until 1:05, then S-ATT; S-HAND at 1:15.
S6	Re-engages	S-TALK with S5 until 1:05, then S-NOTE.

Check in the TeachLens outputs

- Instructor camera: posture sitting/unknown, then standing at 1:00; motion still, then low/medium.
- Student camera: S2 standing about 2:30-2:40; hand raises stay sitting.

Watch for

- The re-engagement at 1:05 should be visible: phones away, sitting up, turning to face the front.

Instructor movement and posture. Standing, walking, pacing, sitting, board writing, leaving, roaming, and legs cut off by the frame.

B1 Standing-still lecture

Priority 2 | Teaching: Effective (static) | Class: Mostly attentive | Length: 3:00

Purpose. The simplest instructor case: one spot, upright, small gestures. A reference for posture standing and still motion.

Cast. 6 students. With 4, drop S5 and S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-1:30	T-STAND	Centre front, small hand gestures only.
1:30-1:45	T-WRITE	Turn and write one line.
1:45-3:00	T-STAND	Back to the same spot.

Students

Seat	Role	What to do (codes and times)
S1	Note-taker	S-NOTE throughout.
S2	Attentive	S-ATT throughout.
S3	Note-taker	S-NOTE throughout.
S4	Late phone	S-ATT; S-PHONE from 2:00.
S5	Attentive	S-ATT throughout.
S6	Restless	S-ATT; S-FIDGET 1:00-1:20.

Check in the TeachLens outputs

- Instructor camera: posture standing (full legs if the feet are in frame); motion still/low; the hip path stays at one spot.

Watch for

- Stand where the whole body (or at least the knees) is visible.

B2 Walking lecture with a side exit

Priority 1: film first | Teaching: Effective (mobile) | Class: Mostly attentive | Length: 3:00

Purpose. Walking speed, pauses, the instructor's path, and one deliberate exit and return through the side of the instructor camera's frame.

Cast. 6 students. With 4, drop S5 and S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-2:00	T-WALK	Walk slowly left and right along the front (about 5 passes), pausing 5 s at each end.
2:00	T-EXIT	Walk out through the side of the frame (door side).
2:00-2:10	-	Stay out of the frame.
2:10-3:00	T-WALK	Come back in and continue walking and pausing.

Students

Seat	Role	What to do (codes and times)
S1	Follows	S-ATT, follows the instructor with the head.
S2	Note-taker	S-NOTE throughout.
S3	Follows	S-ATT throughout.
S4	Follows	S-ATT; S-LOOK 2:00-2:10 toward the door.
S5	Distracted	S-ATT; S-PHONE 1:20-1:40.
S6	Follows	S-ATT throughout.

Check in the TeachLens outputs

- Instructor camera: motion low/medium while walking, still/low at the pauses; one apparent exit about 2:00 and one entry about 2:10 at the side edge; presence not detected 2:00-2:10; probably a new tracker ID after the return.
- With `--path teacher`, the review video shows the walking trail.

Watch for

- Exit through the left or right edge, not the bottom of the frame.

B3 Pacing and large gestures

Priority 2 | Teaching: Nervous / over-animated | Class: Mostly attentive | Length: 2:30

Purpose. The high end of instructor motion: fast pacing, sweeping gestures, frequent turns.

Cast. 6 students. With 4, drop S5 and S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-2:30	T-PACE, T-GEST	Pace quickly, sweep the arms, turn often; sometimes step toward and away from the camera.

Students

Seat	Role	What to do (codes and times)
S1	Attentive	S-ATT throughout.
S2	Attentive	S-ATT throughout.
S3	Amused	S-ATT; S-TALK with S4 1:00-1:15 (whisper, smile).
S4	Amused	S-ATT; S-TALK with S3 1:00-1:15.
S5	Attentive	S-ATT throughout.
S6	Note-taker	S-NOTE throughout.

Check in the TeachLens outputs

- Instructor camera: motion medium/high; posture often unknown mid-stride (bent knees), standing at turns.

Watch for

- Keep pacing inside the frame; the point is speed, not exits.

B4 Seated instructor and posture changes

Priority 1: film first | Teaching: Mixed | Class: Mostly attentive | Length: 3:00

Purpose. Instructor sitting, standing, and perching, with exact transition times. Film **two takes**: take 1 seated behind the desk (legs hidden), take 2 on a chair beside the desk (legs visible).

Cast. 6 students. With 4, drop S5 and S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-1:15	T-SIT	Sit and teach from the desk (take 2: chair beside the desk).
1:15	-	Stand up and walk to the board.
1:30-2:15	T-STAND, T-GEST	Explain at the board.
2:15-2:40	T-LEAN	Perch on the front edge of the desk.
2:40-3:00	T-SIT	Sit back in the chair.

Students

Seat	Role	What to do (codes and times)
S1	Attentive	S-ATT throughout.
S2	Note-taker	S-NOTE throughout.
S3	Attentive	S-ATT throughout.
S4	Looks away	S-LOOK 0:30-1:00, then S-ATT.
S5	Note-taker	S-NOTE throughout.
S6	Phone	S-ATT; S-PHONE 2:20-2:50.

Check in the TeachLens outputs

- Take 1: posture unknown or partial while seated behind the desk (true: sitting); standing at the board.
- Take 2: posture sitting from the legs while seated beside the desk.
- Perching (2:15-2:40) is ambiguous: unknown is the correct, cautious result.

Watch for

- Log the transitions at 1:15, 2:15, and 2:40 to the second.

B5 Board-writing lesson

Priority 2 | Teaching: Mixed | Class: Copying; one opportunist | Length: 3:00

Purpose. Long periods with the instructor's back turned. Heads down here mean copying (on task), while one student uses the phone only when the instructor cannot see.

Cast. 6 students. With 4, drop S5 and S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-0:45	T-WRITE	Write with the back to the class.
0:45-1:05	T-GEST	Turn and explain.
1:05-1:40	T-WRITE	Write.
1:40-2:00	T-GEST	Explain.
2:00-2:40	T-WRITE	Write.
2:40-3:00	T-ASK	Turn and ask a question.

Students

Seat	Role	What to do (codes and times)
S1	Copier	S-NOTE during writing, S-ATT during explanations.
S2	Opportunist	**S-PHONE only while the instructor writes**; hides it when they turn.
S3	Copier	Same as S1.
S4	Copier	Same as S1; S-HAND at 2:45.
S5	Attentive	S-ATT throughout; ignores S6.
S6	Whisperer	S-TALK toward S5 during writing periods; S-ATT otherwise.

Check in the TeachLens outputs

- Instructor camera: presence visible even with the back turned; face landmarks not visible while writing; motion low.
- Student camera: everyone sitting. Ground truth separates head-down copying from phone use.

Watch for

- The phone switch must follow the instructor's turns, which is what makes it realistic.

B6 Near and far: legs leaving the frame

Priority 1: film first | Teaching: n/a (camera test) | Class: Attentive | Length: 2:30

Purpose. Checks the new handling of an instructor close to the camera: feet, knees, and hips leave the bottom of the frame without the instructor leaving the room.

Cast. Any number of students (they just sit attentively).

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-0:35	T-STAND	Stand at the back of the teaching area, whole body visible.
0:35-0:50	T-NEAR	Walk slowly toward the instructor camera until only the waist up is visible.
0:50-1:15	T-STAND	Stay close, talking (waist-up).
1:15-1:30	T-WALK	Step back until the whole body is visible.
1:30-2:00	T-NEAR	Repeat on the left side of the frame.
2:00-2:30	T-NEAR	Repeat on the right side of the frame.

Students

Seat	Role	What to do (codes and times)
All	Attentive	S-ATT or S-NOTE throughout; the student camera still records.

Check in the TeachLens outputs

- Posture: standing (full legs) when far; standing (partial) while only the knees are visible; unknown when waist-up.
- **No** apparent entries or exits while the instructor is cut off at the bottom of the frame.
- The overlay shows "cut off"; with calibrated zones, the zone is unknown while cut off; joints placed outside the image are dropped.

Watch for

- Stay inside the left and right edges; only the bottom edge should cut the body.

B7 Instructor walks among the students

Priority 2 | Teaching: Effective (circulating) | Class: Working | Length: 3:00

Purpose. Documents a known limitation: the instructor leaves the instructor camera and appears in the student camera, where every person is assumed to be a student.

Cast. 6 students. With 4, drop S5 and S6 (S4 asks the question instead of S5).

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-0:40	T-STAND	Give a task at the front.
0:40-1:40	T-ROAM	Walk along the aisle; stop at S2's desk (about 0:55) and S5's desk (about 1:10).
1:40-2:00	T-WALK	Return to the front.
2:00-3:00	T-STAND, T-GEST	Explain the answer.

Students

Seat	Role	What to do (codes and times)
S1	Working	S-NOTE throughout.
S2	Hides phone	S-PHONE until the instructor approaches (about 0:50), then hides it and works.
S3	Working	S-NOTE throughout.
S4	Working	S-NOTE; S-LOOK 1:45-2:00.
S5	Asks for help	S-HAND at 1:05; the instructor bends down to help.
S6	Working	S-LOOK, then S-NOTE from 0:45.

Check in the TeachLens outputs

- Instructor camera: presence not detected about 0:40-1:40 (true: in the room but out of view, so not detected does not mean absent).
- Student camera: the instructor is counted as a student while visible (count = cast + 1). Log those times so the error can be measured.

Watch for

- This take is about the limitation; do not avoid the student camera.

B8 Instructor leaves the room

Priority 1: film first | Teaching: n/a | Class: Off task while unsupervised | Length: 3:00

Purpose. Instructor presence, exit and entry, and how the class changes when nobody is watching.

Cast. 6 students. With 4, drop S5 and S6; S2 does S5's stretch at 1:00-1:10, then goes back to the phone.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-0:40	T-STAND	Assign a task.
0:40	T-EXIT	Walk out through the door (side of the frame).
0:40-1:40	-	Out of the room.
1:40	-	Come back in and walk to the front.
1:40-3:00	T-WALK, T-STAND	Check the work verbally.

Students

Seat	Role	What to do (codes and times)
S1	Keeps working	S-ATT, then S-NOTE throughout.
S2	Phone	S-ATT; S-PHONE 0:45-1:40.
S3	Chats	S-ATT; S-TALK with S4 0:45-1:40, laughing.
S4	Chats	S-ATT; S-TALK with S3 0:45-1:40.
S5	Stretches	S-ATT; **S-STAND 1:00-1:10** stands and stretches in place, sits.
S6	Visits	S-ATT; 1:05 walks over to S4 and S5 (out of the seat); hurries back to the seat at 1:35.

Check in the TeachLens outputs

- Instructor camera: apparent exit about 0:40, presence not detected 0:40-1:40, apparent entry about 1:40.
- Student camera: S5 standing 1:00-1:10; S6 posture unknown while away from the seat (true: standing and walking).

Watch for

- The door must be at a side of the instructor camera's frame.

Student events. Standing answers, arrivals and departures, pair work, silent work, presentations, and media.

C1 Question time: hands and standing answers

Priority 1: film first | Teaching: Effective (interactive) | Class: Attentive | Length: 3:30

Purpose. The main test of student sitting/standing: precise stand-up moments mixed with hand raises that must stay sitting.

Cast. 6 students. With 4: S2 stands for Q5 instead of S6, and S5's phone moment is dropped.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-3:30	T-STAND, T-ASK	Teach standing at the front; ask 5 questions at 0:20, 0:55, 1:30, 2:05, and 2:40; point to the answering student each time.

Questions and answers

Question	Hands raised (S-HAND, 3-5 s)	Answer
Q1 0:20	S1, S3	**S1 stands 0:25-0:33**
Q2 0:55	S2, S4, S5	**S4 stands 1:00-1:08**
Q3 1:30	S3	**S3 stands 1:35-1:43**
Q4 2:05	S2, S6	S2 answers **seated** (hand only)
Q5 2:40	S5, S6	**S6 stands 2:45-2:53**

Students

Seat	Role	What to do (codes and times)
Others	Attentive	S-ATT or S-NOTE between questions; S5 S-PHONE 1:45-2:00.

Check in the TeachLens outputs

- Student camera: standing intervals match the four stand-ups (allow about 1 s: TeachLens checks one frame per second); every other moment is sitting (or unknown at track start).
- Hand raises, including Q4's seated answer, stay sitting.
- The `shoulder_rise` events give the raw rise behind each label for threshold tuning.

Watch for

- Stand up straight in place; sit down fully before the next question.
- Keep both shoulders visible to the student camera when standing.

C2 Late arrival, restroom break, early leave

Priority 1: film first | Teaching: Effective | Class: Mostly attentive | Length: 3:30

Purpose. Visible student count, apparent entries and exits, and tracker IDs when people come and go.

Cast. 6 students; S6 waits outside at the start. With 4: S4 arrives late, S3 takes the restroom break, S2 leaves early.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-3:30	T-STAND, T-WALK	Lecture normally.
0:45	-	Pause, look at the door, gesture S6 to a seat.

Students

Seat	Role	What to do (codes and times)
S1	Note-taker	S-NOTE throughout.
S2	Leaves early	S-ATT; at 2:00 packs the bag, **S-LEAVE 2:05** for good.
S3	Attentive	S-ATT throughout.
S4	Restroom	S-ATT; **S-LEAVE 1:10**; returns 2:30, seated by 2:40.
S5	Attentive	S-ATT; S-PHONE 1:30-1:45.
S6	Late	**S-LATE 0:45**: enters through the door, seated by 0:55.

Check in the TeachLens outputs

- True count: 5, then 6 at 0:55, 5 at 1:15, 4 at 2:05, 5 at 2:40. The visible_student_count should follow with about 1 s delay.
- Apparent entry and exit events at the side edge for each arrival and departure; S4 probably gets a new tracker ID on return (IDs are not people).
- S6 walks in standing: the shoulder-rise rule assumes students start seated, so check S6's first seconds.

Watch for

- The door must be at a side of the student camera's frame, not at the bottom.

C3 Pair work: on task and off task

Priority 2 | Teaching: Effective (activity) | Class: Mixed (4 on, 2 off) | Length: 3:00

Purpose. Students turn toward each other, so shoulders overlap and orientation changes. The ground truth separates on-task talk from off-task talk.

Cast. Pairs by column: S1+S4, S2+S5, S3+S6. With 4: S1+S4 on task, S2+S3 off task.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-0:40	T-STAND	Assign a pair task.
0:40-2:30	T-SIT	Watch from the desk (stay in the instructor camera's view).
2:30-3:00	T-ASK	Ask the pairs to report; S1 stands to report.

Students

Seat	Role	What to do (codes and times)
S1 + S4	On task	S-GROUP 0:40-2:30; **S1 S-STAND 2:35-2:45** to report.
S2 + S5	On task	S-GROUP 0:40-2:30.
S3 + S6	Off task	S-TALK and laughing; S3 turns around and shows S6 a phone (S-PHONE) 1:20-1:40.

Check in the TeachLens outputs

- Student camera: sitting; occlusion flags while partners lean together; posture may be unknown when a turned student hides a shoulder; S1 standing at 2:35.

Watch for

- Turn the whole torso toward the partner; that is what orientation indicators will need later.

C4 Silent individual work (quiz)

Priority 2 | Teaching: n/a (supervision) | Class: On task, heads down | Length: 3:00

Purpose. Very low motion with heads down that is on task: the counterpart to phone use in A2 and A4.

Cast. 6 students. With 4, drop S5 and S6.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-2:00	T-SIT	Sit at the desk watching the class (papers are already on the desks).
2:00-2:40	T-WALK	Walk slowly along the front.
2:40-3:00	T-STAND	Announce "one minute left".

Students

Seat	Role	What to do (codes and times)
S1	Working	S-NOTE throughout.
S2	Stuck	S-NOTE; S-FIDGET 0:50-1:30 (looks around, taps the pen).
S3	Working	S-NOTE throughout.
S4	Stretch	S-NOTE; **S-STAND 2:20-2:28** stretch in place, sit.
S5	Finishes early	S-NOTE until 2:00, then S-PHONE.
S6	Question	S-NOTE; S-HAND at 1:10; the instructor answers from the desk.

Check in the TeachLens outputs

- Both cameras: mostly still motion; students sitting except S4 at 2:20.

Watch for

- Heads down while writing should look the same as in note-taking, so the ground truth carries the meaning.

C5 Student presentation (two people in front)

Priority 2 | Teaching: n/a | Class: Attentive | Length: 3:00

Purpose. Two people in the instructor camera's view, which the camera-role rule must treat cautiously.

Cast. 6 students. With 4: S2 presents and S4 asks the question.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-0:30	T-STAND	Introduce the presenter.
0:30-2:00	T-SIT	Sit at the side, still in the instructor camera's view.
2:00-3:00	T-STAND	Resume at the front.

Students

Seat	Role	What to do (codes and times)
S2	Presenter	0:30 stands, walks to the front; presents at the board 0:40-2:00; back in the seat by 2:05.
S1	Attentive	S-ATT throughout.
S3	Note-taker	S-NOTE throughout.
S4	Phone	S-ATT; S-PHONE 1:00-1:30.
S5	Attentive	S-ATT throughout.
S6	Question	S-ATT; S-HAND at 1:50, asks S2 a question.

Check in the TeachLens outputs

- Instructor camera: roles unknown and teacher presence samples unknown while two people are visible (the cautious, correct behaviour).
- Student camera: the count drops by one while S2 is away from the seat.

Watch for

- S2 should leave the student camera's view through a side edge.

C6 Watching a video

Priority 2 | Teaching: n/a (media) | Class: Mixed | Length: 3:00

Purpose. Students look at a screen instead of the instructor; a still instructor at the side.

Cast. 6 students. With 4, drop S5 and S6 (S4 still sleeps).

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-0:30	T-STAND	Introduce the video.
0:30-2:30	T-SCREEN	Stand still beside the screen.
2:30-3:00	T-ASK	Return to the centre, ask a question; S1 raises a hand.

Students

Seat	Role	What to do (codes and times)
S1	Watching	S-ATT toward the screen; S-HAND at 2:35.
S2	Watching	S-ATT toward the screen.
S3	Notes	S-NOTE throughout.
S4	Dozes	S-ATT; S-SLEEP from 1:15.
S5	Watching	S-ATT toward the screen.
S6	Phone	S-ATT; S-PHONE 1:00-2:00.

Check in the TeachLens outputs

- Instructor camera: standing, still. Student camera: sitting, low motion.
- For future orientation indicators: facing the screen is attentive here, not "looking away".

Watch for

- Place the screen so that facing it is clearly different from facing the instructor.

D1 Occlusion and crowding

Priority 2 | Teaching: Effective | Class: Mostly attentive | Length: 2:30

Purpose. Hidden shoulders, overlapping students, and someone crossing in front of others: the hard cases for tracking and posture.

Cast. 6 students recommended (with 4: S4 sits behind S2 instead of S5).

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-2:30	T-STAND, T-WALK	Teach normally (instructor camera only).

Students

Seat	Role	What to do (codes and times)
S5	Hidden	Sits directly behind S2 (as seen from the student camera) 0:15-1:00; moves to the normal staggered seat at 1:00.
S1	Leans over	S-ATT; leans across to S2's desk to read notes 1:15-1:30.
S3	Crosses	1:45 stands, walks across in front of S1 and S2 to the left wall, returns and sits by 2:10.
S2	Attentive	S-ATT throughout.
S4	Restless	S-FIDGET throughout.
S6	Attentive	S-ATT throughout.

Check in the TeachLens outputs

- possible_occlusion flags; S5 posture unknown while hidden; tracker ID switches are likely when S3 crosses. Compare tracker ID values with the true number of people.

Watch for

- Crossing must happen between the camera and the desks, slowly enough to be safe.

D2 Off-task disturbance and recovery

Priority 2 | Teaching: Effective (manages the class) | Class: Disrupted, then recovers | Length: 3:00

Purpose. A noisy, high-motion moment in the class, the instructor stepping in, and the class settling.

Cast. 6 students. With 4: S2 tosses the paper ball to S1 instead of S6 to S5.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-1:10	T-STAND	Lecture.
1:10-1:30	T-ROAM	Stop, walk toward S3 and S4, speak to them.
1:30-1:45	T-WALK	Return to the front.
1:45-3:00	T-STAND, T-GEST	Resume.

Students

Seat	Role	What to do (codes and times)
S3	Disturbs	S-TALK loudly with S4 0:40-1:20 (laughing, turning); S-ATT after 1:30.
S4	Disturbs	Same as S3.
S6	Tosses	1:00 tosses a soft paper ball to S5; S-NOTE after 1:30.
S5	Target	S-ATT; reacts at 1:00 without tossing back.
S2	Relapse	S-ATT; S-PHONE quietly 2:20-2:40.
S1	Attentive	S-ATT throughout.

Check in the TeachLens outputs

- Student camera: motion bursts 0:40-1:20; the instructor is counted as a student while near the desks (limitation, as in B7).
- Instructor camera: presence not detected 1:10-1:30.
- TeachLens reports observable proxies only; it does not label behaviour as misconduct.

Watch for

- Keep it safe: a soft paper ball only, nothing thrown at faces.

N1 Natural, unscripted mini-lesson

Priority 1: film first | Teaching: Natural | Class: Natural | Length: 5:00

Purpose. Scripted acting tends to be exaggerated. One natural lesson per session shows how real behaviour looks to the model.

Cast. Any 4-6 students.

Standard opening: 0:00 clap and take sign facing both cameras; 0:00-0:15 everyone seated and settled, instructor at the start mark. End: hold positions 5 s, then clap.

Instructor

Time	Code	Action
0:15-5:00	-	Teach a real 5-minute mini-lesson on one of the topics; move as you naturally would.

Students

Seat	Role	What to do (codes and times)
All	Natural	No assigned roles. Behave as you would in a real class (off-task moments are fine). The logger records what actually happened afterwards from the video.

Check in the TeachLens outputs

- Use this take for realism checks and for the most honest accuracy numbers.

Watch for

- Do not tell students what to do; log it afterwards.

Calibration clips. Short clips that give exact ground truth for tuning and zones.

K1 Empty room

Priority 1: film first | Teaching: n/a | Class: n/a | Length: 0:30

Purpose. Background frames for zone calibration, and a false-positive check (no people, no tracks).

Cast. Nobody in either camera's view.

Instructor

Time	Code	Action
0:00-0:30	-	Both cameras record the empty room under the same lighting as the takes.

Check in the TeachLens outputs

- Student count 0; teacher presence not detected; no tracks.
- Use a frame from this clip to draw zone polygons (copy `configs/zones.example.yaml`).

Watch for

- Record it again whenever a camera or the furniture is moved.

K2 Student sit/stand calibration

Priority 1: film first | Teaching: n/a | Class: n/a | Length: 2:30

Purpose. Labelled stand-ups and in-seat movements for tuning the shoulder-rise thresholds (standing 0.7, sitting 0.35 shoulder widths).

Cast. All students; the instructor stays out of both views.

Instructor

Time	Code	Action
0:00-0:20	-	Everyone seated and settled.

Calibration sequence

Time	Who	Action
0:20-0:25	S1	Stand in place, hold, sit
0:35-0:40	S2	Stand in place, hold, sit
0:50-0:55	S3	Stand in place, hold, sit
1:05-1:10	S4	Stand in place, hold, sit
1:20-1:25	S5	Stand in place, hold, sit
1:35-1:40	S6	Stand in place, hold, sit
1:50-1:55	All	Stand together, hold, sit
2:00-2:05	All	Lean forward on the desk (must stay sitting)
2:08-2:13	All	Lean back in the chair (sitting)
2:16-2:21	All	Raise a hand high (sitting)
2:24-2:29	All	Head down on the desk (sitting)

Check in the TeachLens outputs

- The `shoulder_rise` events show the rise for each moment: stand-ups should be well above 0.7, and in-seat movements below 0.35. Adjust `analysis.shoulder_rise_*` in a local override if they are not.

Watch for

- Use the same seats and chairs as in the scenario takes.

K3 Instructor posture and position grid

Priority 1: film first | Teaching: n/a | Class: n/a | Length: 2:30

Purpose. Instructor posture with full legs, hidden legs, and cut-off legs; positions for teaching-area zones.

Cast. Instructor only; students may sit in place.

Instructor

Time	Code	Action
0:15-0:30	T-STAND	5 s each at board-left, board-centre, and board-right.
0:30-1:10	T-WALK	Walk slowly around the edge of the teaching area (the zone corners).
1:10-1:25	T-SIT	Sit behind the desk.
1:25-1:35	T-STAND	Stand behind the desk.
1:35-1:50	T-SIT	Sit on a chair with the legs visible.
1:50-2:10	T-NEAR	Walk toward the camera until waist-up.
2:10-2:30	T-WALK	Step back to full body.

Check in the TeachLens outputs

- Posture: standing (full legs) at the board; sitting from the legs on the open chair; unknown or partial behind the desk and when cut off.

Watch for

- Keep the whole teaching area inside the frame when setting up camera B.

6. Ground-truth log

One person who is not acting keeps this log for every take, live or afterwards from the video. Times are minutes:seconds from the start clap; also write the clap's time in each video file and the wall-clock time of the clap, so the log can be converted to video time and to the real times in the TeachLens output. Give stand-ups and sit-downs to the nearest half second. Log deviations from the script (a missed cue, a late stand-up) and anything unscripted (someone walks past the door). A spreadsheet with the same columns is easiest to compare with the JSONL later.

Take ID: _____ Date: _____ Students present: _____ Rotation: none / take-2 swap Clap at (student video): _____ (instructor video): _____

Person	Code	Start	End	Posture	In seat?	Visible to	Notes

Person: T or S1-S6. Posture: sitting / standing / walking. In seat: yes / no. Visible to: A (student camera), B (instructor camera), both, neither.

7. Running TeachLens and comparing results

File names

Name each file `sgp_<scenario>_t<take>_<camera>.mp4`, for example `sgp_A1_t1_student.mp4` and `sgp_A1_t1_instructor.mp4`, and use the same text (without `.mp4`) as the video ID. Video IDs may contain only letters, digits, hyphens, and underscores.

Commands (from the repository root, TeachLens environment)

```
python scripts/analyze.py --video data/raw_videos/sgp_A1_t1_student.mp4 --video-id sgp_A1_t1_student --camera
student_view
python scripts/analyze.py --video data/raw_videos/sgp_A1_t1_instructor.mp4 --video-id sgp_A1_t1_instructor --
camera instructor_view --path teacher
```

Your own recordings need **no** `--local-config`: the defaults now assume static cameras. Only the downloaded web clips that pan and zoom need `--local-config configs/web_sample_cameras.yaml`. Add `--live` to watch the overlays while it runs. Each run writes the JSONL evidence, a JSON summary, and an annotated review MP4 under `outputs/`.

Comparing with the log

- **Find each student's tracks.** Tracker numbers are anonymous and can change, so match them to seats by position: the `box_center` of a student's `position` events stays at their seat. Watch the review MP4 to confirm.
- **Real time.** Every event also has `start_real_time` and `end_real_time`: the wall-clock time from the phone's recording-time metadata, accurate to about 1 s. Use them to line up the two cameras' results or to compare with a log kept by the clock; use the clap for anything finer than a second.
- **Posture.** Compare `sitting_standing` intervals with the log's stand-up and sit-down times. Count a stand-up as found if a standing interval overlaps it. Report how much time was unknown separately; unknown is not a mistake, but it is not a correct answer either.
- **Count and presence.** Compare `visible_student_count` and `teacher_presence_sample` with who was really in view. Remember the 1 s confirmation delay.
- **Entries and exits.** Every true arrival or departure across a side edge should have an apparent entry or exit near that time; extra ones usually mean the tracker lost someone.
- **Tuning.** If stand-ups are missed or in-seat movements look like standing, use the K2 clip's `shoulder_rise` values to adjust `analysis.shoulder_rise_standing_min` and `shoulder_rise_sitting_max` in a `configs/*.local.yaml` override, and rerun with a new video ID.
- Keep a separate set of takes (for example, all take-2 videos) that you never tune on, and report final accuracy on those only.

> **What the results can and cannot claim**
>
> Results from these takes measure how well TeachLens detects **acted, observable behaviour in your room with your cameras**. They are not a measure of real teaching quality or real student attention, and acted behaviour is cleaner than real behaviour; the natural N1 takes show how large that gap is.